MIL-A-48313(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

ACCELO-SWITCH ASSEMBLY

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for one type of electircal switch entitled Acceloswitch used on the Honest John atomic warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-8484	- Seals and Seal Testing Procedures (For Electronic Inclosure).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

EL-8842560	- Equipment Lists.
8846864	- Acceloswitch.
8846868	- Slider Assembly.

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8846873 - Body and Monitor Assembly.
8846894 - Capsule and Orifice Plate Assembly.
8846896 - Orifice Plate.
8846897 - Capsule Timer.
8846898 - Shaft Timer.
8846899 - Timer Cover.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The acceloswitch shall comply with all requirements specified on the Drawing (dwg) 8846864, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.5 Circuit resistance.

3.5.1 Safe condition. The resistance between terminals B1 and B2 shall not exceed 3.5×10^{-2} ohms when tested as specified in 4.4.1.1.

3.5.2 Operated condition. The resistance between terminals A1 and A2 shall not exceed 2.5×10^{-2} ohms and between terminals C1 and C2 shall not exceed 2.5×10^{-2} ohms when tested as specified in 4.4.1.1.

3.6 Insulation resistance.

3.6.1 Safe condition. The insulation resistance between each terminal and all other terminals except B2 and between all terminals and the case shall not be less than 40×10^{6} ohms when tested at a potential of 500 $\pm$ 50 volts direct current (vdc) as specified in 4.4.2.

3.6.2 Operated condition. The insulation resistance between each terminal and all other terminals except A2 and C2 and between each terminal and the case will not be less then 40×10^{6} ohms when tested at a potential of 500 $\pm$ 50 vdc as specified in 4.4.2.

3.7 Operation time stability.

3.7.1 Thirty days. The acceloswitch shall have complied with the requirements of 3.5.1 and 3.6.1, prior to post reset conditioning and shall comply with test 2 Table I as specified in 4.4.4. The acceloswitch shall then comply with the requirements of 3.5.2, 3.6.2 and 3.10 in the order specified. Following post reset conditioning, the accelo-

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switch shall be stored, terminals upwards, at 80 ± 20 degrees Fahrenheit (F°) in the safe condition for a minimum of 30 days. The acceloswitch shall then be placed in the centrifuge without post reset conditioning and operated in accordance with test 2, of Table I as specified in 4.4.4. The operating time shall be recorded. The 30 day operating time shall not differ from the zero day operating time by more than ± 2 percent. Testing shall be as specified in 4.4.3.1. All switches falling within the limit of ± 3 percent but greater than ± 2 percent shall be accepted by the Government, for an additional 30 day operational time stability test, where the 60 day operation time shall not differ from the zero day operating time by more than ± 3 percent.

3.7.2 Six months. The acceloswitch shall have compiled with the requirements of 3.5.1 and 3.6.1, immediately prior to post reset conditioning and shall comply with test 2 of Table I in the order specified. Test 2 of Table I shall be conducted as specified in 4.4.4. The acceloswitch shall then comply with the requirements of 3.5.2, 3.6.2 and 3.10. Following post reset conditioning, the acceloswitch shall be stored, terminals upward, at 80 ± 20°F in the safe condition for a minimum of 180 days. The acceloswitch shall then be placed in the centrifuge and operated in accordance with test 2 of Table I as specified in 4.4.4. The acceloswitch operating time shall not differ from the zero-day operating time by more than ± 3.5 percent or 0.05 seconds whichever is minimum, and no operating time shall occur outside the limits specified for test 2 of Table I. Testing shall be as specified in 4.4.3.2. After completion of the 180-day test and removal of the acceleration field, the acceloswitch shall comply with the requirements of 3.5.2 and 3.6.2.

3.8 Environmental testing. The acceloswitch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The acceloswitch shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.8.1 Acceleration at 80 + 20°F.

3.8.1.1 No-Go. Electrical continuity between terminals B1 and B2 shall not be broken when the acceloswitch is subjected to an acceleration of 6.5 ± 0.5 gravity units (g's) applied for not less than 8 seconds in the direction required for operation (see Figure 1). Testing shall be as specified in 4.4.4.1.

3.8.1.2 Operation-no load.

3.8.1.2.1 Electrical continuity between terminals B1 and B2 shall be broken within the time specified in Table I. In addition, electrical continuity between terminals A1 and A2 and between terminals C1 and C2 shall be established within the time limits specified in Table I when the acceloswitch is accelerated on time in accorance with test

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I and three times each in accordance with tests 2 and 3. Testing shall be as specified in 4.4.4.2. After the acceleration field has been removed, the acceloswitch shall remain in the operated condition and shall comply with the requirements of 3.5.2 and 3.6.2 immediately following the final acceleration in accordance with test 3 only.

Table I

Test	Acceleration (g's)	Time to Operate Min (seconds)	Time to Operate Max	Monitor Opening Time (seconds)
1.	12.0 ± 0.3	--	60.00	No data required
2.	15.0 ± 0.4	2.11	2.53	0.25 max
3.	38.0 ± 0.8	1.30	1.57	0.16 max

3.8.1.3 Operating current load. During the operating acceleration, circuits A1 - A2 and C1 - C2 in parallel shall close and carry a resistive dc load of 20 ± 1 amperes applied for 60 ± 0.40 milliseconds followed by a constant dc load of 5.5 ± 0.3 amperes for a minimum of 22 seconds. During this test, the parallel resistance shall not exceed 1.3 X 10-2 ohms subsequent to a 0.010 second period after closure of the first contact. Testing shall be as specified in 4.4.4.1.3.

3.8.2 Current carrying capability. Each contact in the operated condition, excluding B1 and B2, shall be connected to and shall then carry the load current specified in Figure 2.

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The maximum voltage drop between A1 and A2 and between C1 and C2 shall not exceed 1.3 volts during the 22 second load cycle. Testing shall be as specified in 4.4.4.2.

3.8.3 Reset capability. The acceloswitch shall not reset when an acceleration of 60 g's is applied in the direction opposite to that required for operation for 30 seconds maximum as indicated by discontinuity between terminals B1 and B2. The acceloswitch shall reset from the operated to the safe condition in 3 minutes maximum when an acceleration of less than 200 g's is applied in the direction opposite to that required for operation as indicated by continuity between terminals B1 and B2. Testing shall be as specified in 4.4.4.3.

3.8.3.1 Reset. The acceloswitch shall be accelerated at a "g" sufficient for complete resetting as indicated by continuity between terminals B1 and B2, when accelerated in a direction opposite to that required for arming. Testing shall be as specified in 4.4.4.3.1.

3.8.4 Leakage. The seal shall be equal to grade A of MIL-S-8484. The leakage rate shall not exceed 1×10^{-7} cubic centimeters per second of helium and shall show no evidence of bubbles when tested as specified in 4.4.4.4.

3.8.5 High temperature. Electrical continuity between terminals B1 and B2 shall be broken within the time specified in test 2 of Table I and electrical continuity between terminals A1 and A2 and between C1 and C2 shall be established within the time limits specified in test 2 of Table I when the acceloswitch is tested one time at 160 ± 5°F as specified in 4.4.4.5. After the acceleration field has been removed, the acceloswitch shall remain in the operated condition and shall comply with the requirements of 3.5.2 and 3.6.2 when tested within 2 minutes after removal of the acceloswitch from the temperature chamber.

3.8.6 Low temperature. The acceloswitch shall comply with the requirements of 3.8.5 except that the temperature shall be -40 ± 5°F.

3.8.7 Impact shock.

3.8.7.1 Safe condition. The acceloswitch shall withstand 40 impact shocks of 22 ± 2 g's peak amplitude for 10 ± 2 milliseconds at the 10g level applied in the direction to induce operation. After this test, and prior to removal from the shock producing device, the A1 - A2 and C1 - C2 circuits shall show no evidence of closure and the B1 - B2 circuits shall indicate closure. The acceloswitch shall be in the safe condition and shall comply with the requirements of 3.5.1, 3.6.1 and 3.8.5 after testing as specified in 4.4.4.6.1.

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3.8.7.2 Operated condition. The acceloswitch shall withstand three impact shocks of 80 ± 8 g's peak amplitude for a duration of 3 ± 1 millisecond at the base in the direction opposite to that required for operation. During the 5 millisecond interval immediately following the initiation of the shock pulse, the continuity between terminals A1 and A2 and between terminals C1 and C2 shall be maintained. Testing shall be as specified in 4.4.4.6.2.

3.8.8 Vibration.

3.8.8.1 Safe condition. The acceloswitch shall withstand transportation vibration at the frequencies, amplitudes, and for the times specified in Table II without visual evidence of damage or of functional deteriorations. Testing shall be as specified in 4.4.4.7.1. After completion of vibration, the acceloswitch shall be in the safe condition and shall comply with the requirements of 3.5.1, 3.6.1 and 3.8.4.

Table II

Frequency Range Hertz (Hz)	Double Displacement Peak g Level (g's)	Cycle Time to Cover Freq. (minutes) Range	Test Time for each Axis (minutes)
10-20-20	2 ± 0.1	3 ± 0.5	10 ± 0.1
20-50-20	0.036 ± 0.0018	3 ± 0.5	10 ± 0.1
50-500-50	5 ± 0.25	3 ± 0.5	10 ± 0.1

3.8.8.2 Operated condition. The acceloswitch shall withstand flight vibration through a frequency range of 75-1000-75 Hertz (Hz) at a peak of 5 ± 0.25 g's for a period of 3 ± 0.3 minutes per axis followed by a vibration at 450 Hz at a peak of 5 ± 0.25 g's for a period of 2 ± 0.2 minutes per axis. During vibrations, the electrical resistance between terminals A1 and A2 and between C1 and C2 shall not exceed 0.025 ohms and continuity between A1 and A2 and between C1 and C2 shall be maintained when tested as specified in 4.4.4.7.2. After completion of vibration, the acceloswitch shall be in the operated condition and shall show no visual evidence of damage and shall comply with the requirements of 3.5.2, 3.6.2 and 3.8.4.

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3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 50 complete assemblies of each (see dwg 8846864) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

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Table III

First article inspection and tests

Inspection or Test	Units 1-10	11-30	31-40	41-50	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X		X
2. Leakage (see 3.8.4)	X	X	X	X		X
3. Acceleration-No Go (see 3.8.1.1)	X	X	X	X		X
4. Operation-No Load (see 3.8.1.2)	X	X	X	X		X
5. High temperature (see 3.8.5)		X				X
6. Low temperature (see 3.8.6)			X	X		X
7. Impact shock (safe) (see 3.8.7.1)	X				X	
8. Vibration (safe) (see 3.8.8.1)		X			X	
9. Operation-No Load (see 3.8.1.2)	X	X	X	X		X
10. Vibration operated (see 3.8.8.2)		X			X	
11. Current carrying capability (see 3.8.2)	X	X	X	X	X	
12. Reset (see 3.8.3)	X	X	X	X		X
13. Operating current load (see 3.8.1.3)	X	X	X			X
14. Leakage (see 3.8.4)	X	X	X	X		X
15. Impact shock-operated (see 3.8.7.2)				X	X	
16. Leakage (see 3.8.4)				X		X

3 One 15 g run.

4 For operation - No Load Test, required after temperature test.

5 (same comment as note 4).

7 Circuit and insulation resistance tests shall not be performed.

8 Only the Table I tests specified below are required. Units 1-10 and 31-40: 5 units at 15 g's and 5 units at 38 g's. Units 11-30: 10 units at 15 g's and 10 units at 38 g's.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Capsule timer (see dwg 8846879 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Diameter (0.6760 + 0.0005 - 0.000)	Gage
Major:		
	101. Objective evidence relating to material missing or improper (see 3.1)	Visual
	102. Evidence of poor workmanship (see 3.9)	Visual
Minor: None defined.		

4.3.2.1.2 Orifice plate (see dwg 8846896 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Diameter (0.2075 + 0.0005 - 0.000)	Gage
Major:		
	101. Radius (0.002 max)	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
	103. Evidence of poor workmanship (see 3.9)	Visual
Minor: None defined.		

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4.3.2.1.3 Capsule and orifice plate assembly (see dwg 8846894 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Perpendicularity (0.519 ± 0.002)	Gage
2.	Finish on inner surface improper or inadequate	Visual
Major:		
101.	Objective evidence relating to material missing or improper (see 3.1)	Visual
102.	Evidence of poor workmanship (see 3.9)	Visual
Minor: None defined.		

4.3.2.1.4 Timer cover (see dwg 8846899 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Diameter (0.6750 - 0.0005)	Gage
2.	Diameter (0.1840 - 0.0005)	Gage
Major:		
101.	Concentricity (0.6750)	Gage
102.	Concentricity (0.1840)	Gage
103.	Objective evidence relating to material missing or improper (see 3.1)	Visual
104.	Evidence of poor workmanship (see 3.9)	Visual
Minor: None defined.		

4.3.2.1.5 Shaft timer (see dwg 8846898 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Diameter (0.675 - 0.0005)	Gage
2.	Concentricity of the 0.6750 diameter (0.001 TIR)	Gage

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Major:
101. Objective evidence relating to material missing or improper (see 3.1).....Visual
102. Evidence of poor workmanship (see 3.9).....Visual

Minor: None defined.

4.3.2.1.6 Body and monitor assembly (see dwg 8846873 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Improper alignment or contact closure of "B" monitoring contact.....Visual
102. Objective evidence relating to material missing or improper (see 3.1).....Visual
103. Evidence of poor workmanship (see 3.9).....Visual

Minor: None defined.

4.3.2.1.7 Body and monitor assembly (see dwg 8846873 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Location and alignment of all arming contact assemblies.....Gage
102. Objective evidence relating to material missing or improper (see 3.1).....Visual
103. Evidence of poor workmanship (see 3.9).....Visual

Minor: None defined.

4.3.2.1.8 Slider assembly (see dwg 8846868 covering a detail of dwg 8846864).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:
101. Location and alignment of shorting bars.....Gage
102. Objective evidence relating to material missing or improper (see 3.1).....Visual
103. Evidence of poor workmanship (see 3.9).....Visual

Minor: None defined.

4.3.2.1.9 Acceloswitch assembly (see dwg 8846864).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Marking improper or illegible.....Visual
102. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Length of case.....Gage
202. Length of terminal.....Gage
203. Length of pinch-off tube.....Gage
204. Diameter of case.....Gage
205. Terminal configuration improper.....Visual
206. Terminal bent.....Visual
207. Evidence of poor workmanship (see 3.9).....Visual
208. Soldering inadequate.....Visual
209. Serialization missing or improper (see 3.4).....Visual

4.3.2.1.10 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Preservation, packing and packaging not in accordance with MIL-STD-1169.....Visual
102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:
201. Evidence of poor workmanship (see 3.9).....Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Leakage	Major
b. Acceleration no-go	Major
c. Operation time stability	Major
d. Leakage	Major
e. Impact shock (safe)	Major
f. Vibration (safe)	Major
g. Vibration (operated)	Major

Of these tests, e, f and g are considered destructive and samples so tests shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Leakage, acceleration no-go, operation time, stability and leakage. Each acceloswitch in the lot shall be tested in accordance with 4.4.4.4, 4.4.4.4.1 and 4.4.3. If any item fails to comply with 3.8.4, 3.8.1.1 and 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 Impact shock (safe condition), vibration (safe condition) and vibration (operated condition). Three sample acceloswitches, one for each of these tests, shall be selected at random from each lot of 50 acceloswitches, or less, presented for acceptance. The samples shall be tested in accordance with 4.4.4.6.1, 4.4.4.7.1 and 4.4.4.7.2. At the completion of these tests, except acceloswitches subjected to vibration operated test, each sample acceloswitch shall comply with the requirements of 3.8.1.2, test 2 of Table I when tested three times as specified in 4.4.4.1.2. If any sample fails to meet the requirements of 3.8.7.1, 3.8.8.1, 3.8.8.2 and 3.8.1.2.1, the lot represented by the sample in which failure occured shall be rejected.

4.3.3.3 Operation time stability (6 months). One acceloswitch shall be selected from each lot of ten consecutively sealed and accepted acceloswitches for this test (see 4.4.4.1.3). The units selected shall be subjected to two runs in accordance with Table I at 80 ± 20°F in accordance with 4.4.3.2 and no operating time shall occur outside the limits specified for test 2 of Table I. If the above conditions are met, the lot will be considered acceptable. If any sample is not stable or not within acceptable limits, the Government will have the right of exercising the following options if they so desire:

a. All units within the test sample found to be unacceptable shall be reworked by the contractor at no cost to the Government.

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b. The Government will have the right to return to the contractor a percentage of the production lot represented by the tested sample to the percent reject found within the sample lot.

c. In the event the Government elects not to return the defective acceloswitches, the contractor shall be penalized 50 percent of the original contract value of the acceloswitches. At no time will this quantity of acceloswitches for which reinbursement is claimed be in excess of the percentage of failures recorded in the sample test lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8842560). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (stationary). The circuit resistance shall be measured by means of a suitable low electrical resistance measuring instrument which has been connected to the terminals specified. Care must be exercised to avoid breakage of glass seals. A current of 100 milliamperes max shall be used to conduct this test.

4.4.1.1 Circuit resistance (vibrating). The acceloswitch shall be placed in a fixture capable of being rotated to the vertical position, acceloswitch terminals upward, and rotated to three positions 90° apart, starting at position 1 as shown on Figure 1, in the horizontal position, and connected by means of the terminals to a suitable low electrical

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measuring device. The fixture shall then be placed on a suitable low measuring device. The fixture shall then be placed on a vibrator capable of producing 0.005 ± 0.002 inch double amplitude and 60 Hz. The fixture shall be vibrated for duration of test and the circuit resistance shall be measured during this vibration. A current of 100 max milliamperes shall be used to conduct this test.

4.4.2 Insulation resistance (stationary). An insulation resistance tester, from which a reading in ohms can be obtained, shall be used to measure the insulation resistance. The specified potential shall be limited to 5 milliamperes to protect human life. The potential shall be applied until the meter reading becomes steady but the test shall last no longer than 1 minute.

4.4.2.1 Insulation resistance (vibrating). An insulation resistance tester, from which a reading in ohms can be obtained, shall be used to measure the insulation resistance. The specified potential shall be limited to 5 milliamperes to protect human life. The acceloswitch shall be connected to a fixture capable of being rotated to a vertical position, acceloswitch terminals up, and rotated to three positions 90° apart in the horizontal position. The fixture shall be placed on a vibrator capable of producing 0.005 ± 0.002 inch double amplitude and 60 Hz. The fixture shall be vibrated for a minimum of 10 seconds in each position and the potential shall be applied until the meter reading becomes steady, but the test shall last no longer than 1 minute.

4.4.3 Operation time stability.

4.4.3.1 Thirty days. Prior to conducting this test, the acceloswitch, in the safe condition, shall have been tested as specified in 4.4.1 and 4.4.2.1. The acceloswitch shall be tested as specified in 4.4.4.1. The acceloswitch shall be tested, stored and retested to determine compliance with requirements, after which it shall be tested in accordance with 4.4.4.1.2.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Acceleration at 80 + 20°F. Acceleration tests shall be run on a centrifuge using a rotating, shockfree fixture to bring the acceloswitch into proper orientation for operating. For all acceleration tests the direction of rotation of the fixture shall be the same as the direction of rotation of the centrifuge. The major axis of the acceloswitch shall be placed perpendicular to the arm of the centrifuge within 2 degrees when the fixture is in its initial position. The centrifuge shall develop the desired acceleration field within 1 minute after the start of rotation of the centrifuge. The fixture shall rotate the acceloswitch into the acceleration field only after the proper acceleration is attained. At this time the acceloswitch shall be oriented as shown on Figure 1. The major axis shall be parallel to the

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direction of the acceleration within 2 degrees. The fixture shall be periodically calibrated to insure that it rotates from its initial position to its final position in 0.047 to 0.0150 seconds, when the acceleration level is 15 gravity units and in 0.047 to 0.110 seconds when the acceleration is 38 gravity units. The time of the "B" monitoring contacts and the "A" and "C" operational contacts shall be monitored from the initial motion of the swing fixture. Following each single test, or each group of two or more successive tests the acceloswitch shall be tested as specified in 4.4.1, 4.4.2, 4.4.1.1, 4.4.2.1, 4.4.4.3.1, 4.4.1, 4.4.2, 4.4.1.1 and 4.4.2.1, except for the test specified in 4.4.4.1.1 after which the acceloswitch shall be tested as specified in 4.4.4.3.2.

4.4.4.1.1 No-go. The acceloswitch shall be tested as specified in 4.4.4.1.

4.4.4.1.2 Operation no-load. The acceloswitch shall be tested as specified in 4.4.4.1.

4.4.4.1.3 Operating current load. The operating current load shall be run on a centrifuge at the 15 g level using the shockfree fixture specified in 4.4.4.2. A circuit capable of supplying the specified current load shall be applied prior to acceloswitch closure. The voltage drop between circuits A1 - A2 and C1 - C2, in parallel shall be monitored during this test.

4.4.4.2 Current carrying capability. Contacts A1 and A2 shall be connected to a suitable current producing device capable of producing the current time cycle shown on Figure 2. The voltage drop between A1 and A2 shall be monitored during this test. The test shall be repeated using contacts C1 and C2.

4.4.4.3 Reset capability. The acceloswitch shall be mounted on a suitable centrifuge so as to apply an acceleration force within ± 2° of the axis of the acceloswitch and in a direction opposite that required for the arming operation. Following this test the acceloswitch shall be conditioned as specified in 4.4.4.3.2.

4.4.4.3.1 Reset. The acceloswitch shall be tested as specified in 4.4.3.

4.4.4.3.2 Post reset conditioning. The acceloswitch shall be vibrated; terminals upward at a double amplitude of 0.005 ± 0.002 inch at 60 Hz for 4 seconds minimum, after which the terminals of the acceloswitch shall not be oriented below the horizontal plane.

4.4.4.4 Leakage. The sealed acceloswitch shall be tested by utilizing a leak detection method involving the passage of tracer gas from one side of the presumed leak, to the other side. The instrument shall provide a method of determining the leakage rate and the sensitivity of the instrument shall be determined as specified in MIL-S-8484. If no leakage is detected during this test, immerse the switch for 30 seconds

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minimum into a container containing water whose temperature is 175 + 15°F. Minimum leaks will be evidenced by the growth of bubbles at any point. Small bubbles formed by the release of absorbed air shall not be interpreted as leaks. Bubbles indicating leakage shall be cause for rejection.

4.4.4.5 Temperature test. The acceloswitch shall be conditioned in accordance with 4.4.4.3.1 and shall then be stored in a temperature chamber, terminals upward, at the specified temperature, for a minimum of 4 hours or until stabilization (see 6.4) is reached, whichever is longer. The acceloswitch shall then be tested as specified in 4.4.1 and 4.4.2 while in the safe condition. The unit shall then be transferred to a temperature controlled chamber on the centrifuge and assembled in the position required for operation (see Figure 1) and the chamber shall be closed. The testing, transferring, and closing shall be accomplished in not more than 10 minutes. The unit shall be retained and stabilized (see 6.4) in this temperature chamber for a minimum of 1 hour or until stabilization (see 6.4) is reached, whichever is longer at the specified temperature prior to performance of the centrifuge run as specified in 4.4.4.1.2. The chamber temperature shall not change during the centrifuge run by more than 5 degrees. Following this test, the acceloswitch shall be tested as specified in 4.4.1 and 4.4.2 while still at the specified temperature. Insulation resistance failures due to formation of frost on cold acceloswitches shall not be cause for rejection and precautions as necessary for prevention of such indications will be taken.

4.4.4.6 Impact shock .

4.4.4.6.1 Safe condition. The acceloswitch shall be rigidly mounted to a fixture mounted on a suitable shock producing device and shock in the operating direction as shown on Figure 1. Shocks shall be applied at the rate of one shock per minute. The monitor contacts shall be checked for continuity after this test. The acceloswitch shall then be tested as specified in 4.4.1 and 4.4.2 and 4.4.4.4.

4.4.4.6.2 Operated condition. The acceloswitch shall be rigidly mounted to a suitable shock producing device and shocked three times in the direction opposite to that required for operation. Each shock shall be applied for the specified time. The continuity between terminals A1 and A2 and between C1 and C2 shall be monitored during this test with an instrument capable of detecting circuit openings of 20 microseconds minimum duration.

4.4.4.7 Vibration.

4.4.4.7.1 Safe condition. The acceloswitch shall be rigidly mounted to a fixture capable of readily supporting the acceloswitch in each of three mutually perpendicular axes and the fixture shall be mounted on a suitable vibration producing apparatus. The acceloswitch shall be vibrated at the

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frequencies and amplitudes and for the time specified. The monitor contacts shall be checked for continuity after this test. Following completion of the vibration test, the acceloswitch shall be tested as specified in 4.4.1, 4.4.2 and 4.4.4.4.

4.4.4.7.2 Operated condition. Prior to conducting this test, the acceloswitch shall be armed, after which the acceloswitch shall be rigidly mounted to a fixture capable of readily supporting the acceloswitch in each of three mutually perpendicular axes and the fixture shall be mounted on a suitable vibration producing apparatus and vibrated for the time specified. The resistance between terminals A1 and A2 and between C1 and C2 shall be monitored during this test. Also, continuity shall be maintained between A1 and A2 and between C1 and C2 as indicated by monitoring contacts with an instrument capaple of detecting openings of 20 microseconds duration of greater. Following this test, the acceloswitch shall be tested as specified in 4.4.1, 4.4.2 and 4.4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The acceloswitch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The acceloswitch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The acceloswitch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1678 Rev. 3 dated 24 July 1963.

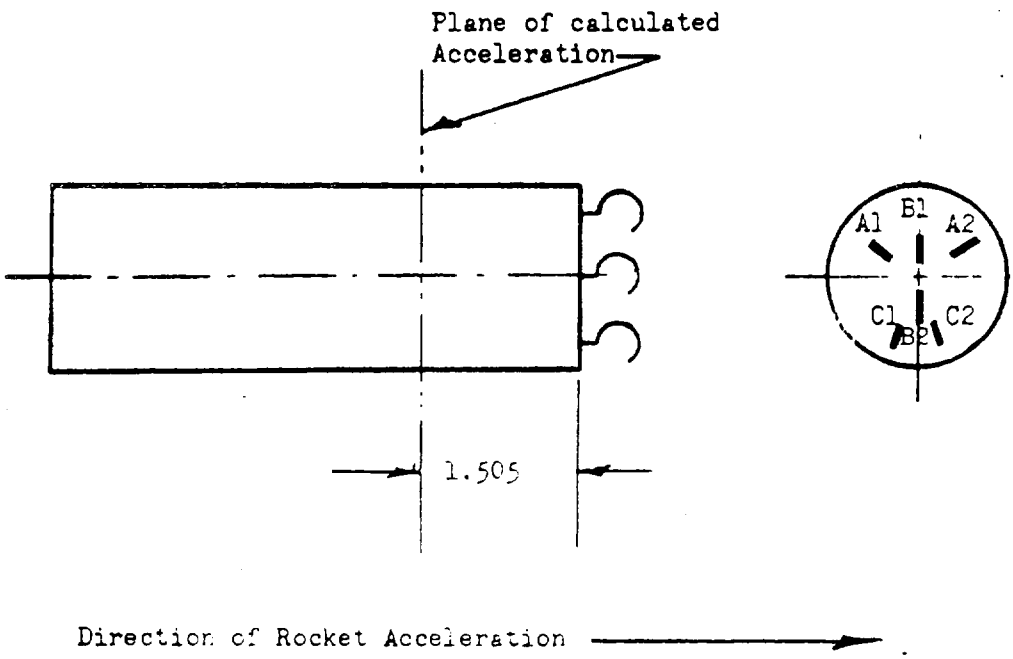
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CENTRIFUGE TESTS

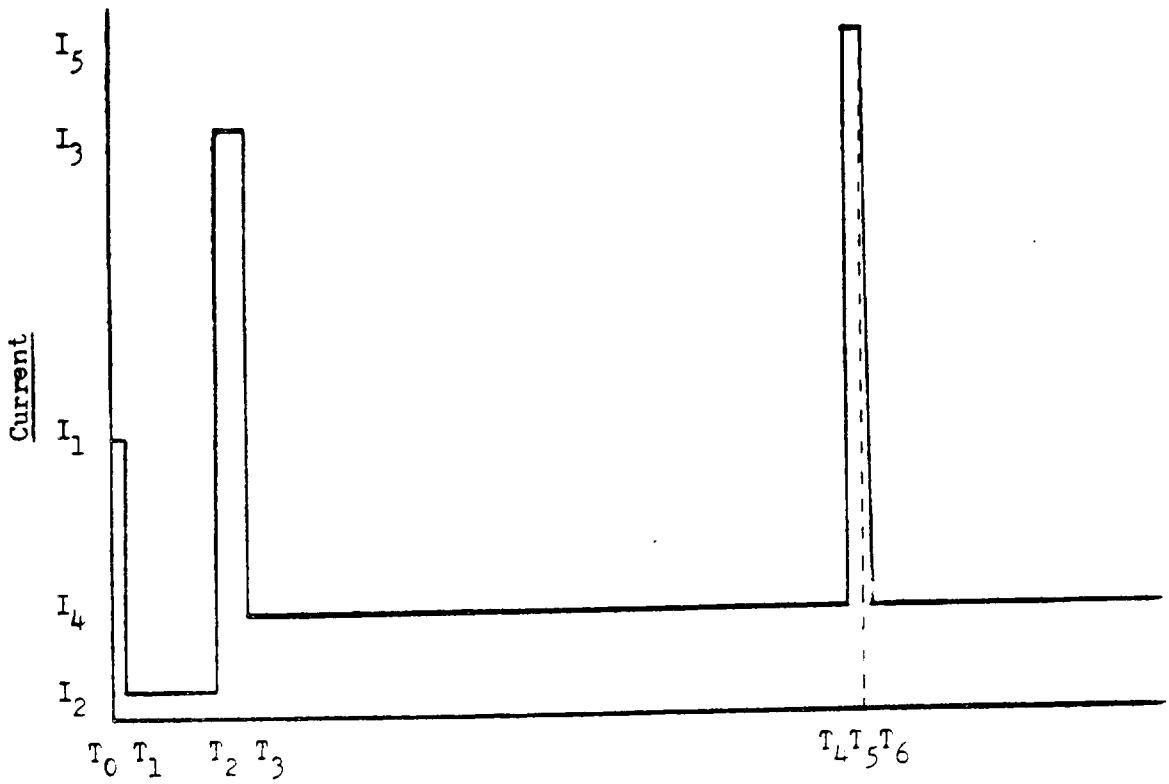


Direct terminals of Acceloswitch toward center of centrifuge to simulate actual operation.

Figure 1

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ACCELOSWITCH-CURRENT CARRYING CAPABILITY



Time

T_0 = Time of current application for current carrying capability (see 3.8.2)
$T_1 - T_0$ = 60 ± 40 Milliseconds
$T_2 - T_0$ = 2 ± 0.5 Seconds
$T_3 - T_2$ = 125 ± 75 Milliseconds
$T_4 - T_0$ = 21 ± 1.0 Seconds
$T_5 - T_4$ = 5 ± 0.4 Milliseconds
$T_6 - T_0$ = 22 Seconds minimum

Current

I_1 = 14.2 ± 0.8
I_2 = 2.0 ± 0.2
I_3 = 35.0 ± 3.0
I_4 = 5.5 ± 0.3
I_5 = 45.0 ± 5.0

Figure 2

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DOCUMENT IDENTIFIER AND TITLE
ACCELO-SWITCH ASSEMBLY

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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